AI Usage DISCLOSURE FORM

1. **Team Details**

Team Name: HealMesh

Project / Product Name: SELF-HEALING ENTERPRISE AUTOMATION MESH

Organization / Institution (if any): R. V. College of Engineering

Submission Date: 8th May 2026

2. **AI Usage Declaration**

Did your team use any Artificial Intelligence (AI) in developing this project? Yes / No

__________Yes__________________

3. **Purpose of AI Usage (Brief Details)**

Idea generation / brainstorming: Architecture mapping, agent workflow design, failure pattern research

Code generation or assistance: n8n JSON scaffolding, Python middleware stubs, Helm/K8s YAML templates, shell script logic

UI / UX design: Minimal (dashboard wireframes only)

Content creation: CONTRACTS.md, HANDOFF.md, README structure, policy documentation

Data analysis: Log parsing, incident payload validation, K8s event correlation

Testing / debugging: Webhook smoke tests, Helm deployment troubleshooting, n8n execution tracing

Other: Prompt engineering, agent behavior tuning, RBAC scope validation

4. **Feature Origin Classification (Please add/delete based on the number of the feature)**

1. Feature Name: n8n Incident Webhook & Validation Workflow

2. Self-Generated/AI-Generated/Both: Both

3. Description (include AI tools/platform Used, prompt used , Output Summary, Modification) :

- AI Tools Used: GitHub Copilot, ChatGPT
- Prompt Used: *"Design an n8n webhook workflow that validates incident JSON against a schema, maps fields to OpenClaw API format, and implements retry + error fallback logic."*
- Output Summary: Baseline n8n JSON with validation nodes, HTTP request stubs, and conditional branching.
- Modification: Manually refactored error handling, added custom JavaScript mappers to align with backend/CONTRACTS.md, disabled placeholder HTTP nodes until middleware was ready, and integrated audit logging via n8n Executions.

2. Feature Name: OpenClaw AI Diagnosis & Remediation Agent

2. Self-Generated/AI-Generated/Both: AI-Generated (Core reasoning) / Both (Integration)

3. Description:

- AI Tools Used: OpenClaw runtime + local Ollama models (Llama 3 / Mistral)
- Prompt Used: *"Act as a Kubernetes SRE. Analyze the provided pod logs and deployment YAML. Identify the root cause of the failure and propose a safe remediation action (PATCH, RESTART, SCALE, or ROLLBACK) in structured JSON. Do not suggest destructive commands."*
- Output Summary: Structured diagnosis with recommended kubectl/helm commands and confidence scores.
- Modification: Team added policy validation layers, human-approval gates, mapped outputs to infra/scripts/remediate.sh action types, and enforced strict JSON schema compliance to prevent malformed cluster mutations.

3. Feature Name:Kubernetes Remediation Execution & RBAC Policies

2. Self-Generated/AI-Generated/Both: Self-Generated

3. Description:

- AI Tools Used: Syntax validation only (Copilot for YAML linting)
- Prompt Used: *"Review this Kubernetes ServiceAccount and RoleBinding for least-privilege access to Deployments and Pods in namespace 'prism'."*
- Output Summary: Validated RBAC manifest structure.
- Modification: Entirely hand-designed by infra team. Implemented namespace isolation, scoped permissions, dry-run flags in remediate.sh, and audit trail logging. No AI-generated execution paths were deployed without manual security review.

4. Feature Name:Policy & Approval Guard (Human-in-the-Loop)

2. Self-Generated/AI-Generated/Both: Both

3. Description:

- AI Tools Used: ChatGPT, n8n community templates
- Prompt Used: *"Create a lightweight approval workflow where n8n pauses execution, sends a Slack/Teams message with Approve/Reject buttons, and resumes based on the response. Include timeout fallback."*
- Output Summary: Prototype sub-workflow with webhook listeners and interactive button handling.
- Modification: Integrated with OpenClaw's policy engine, added timeout fallbacks, manual override controls in frontend, and aligned with security baseline requiring explicit human confirmation for cluster mutations.

5. **Ethical & Compliance Confirmation**

AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I Agree**

6. **Declaration & Sign-Off**

Name of Team Representative: Anwesha Mohapatra

Role: Middleware (Agentic AI + OpenClaw Integration)

Signature: 

Date: 8th May 2026